Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

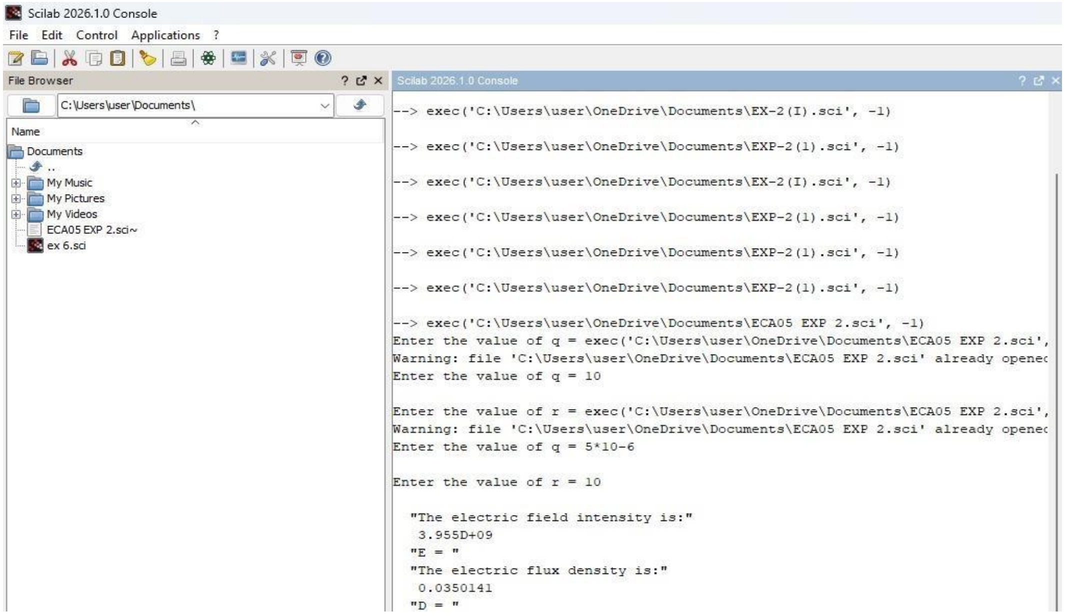
To write program for electric field and electric flux density in sphere using SCILAB. **Software required:**

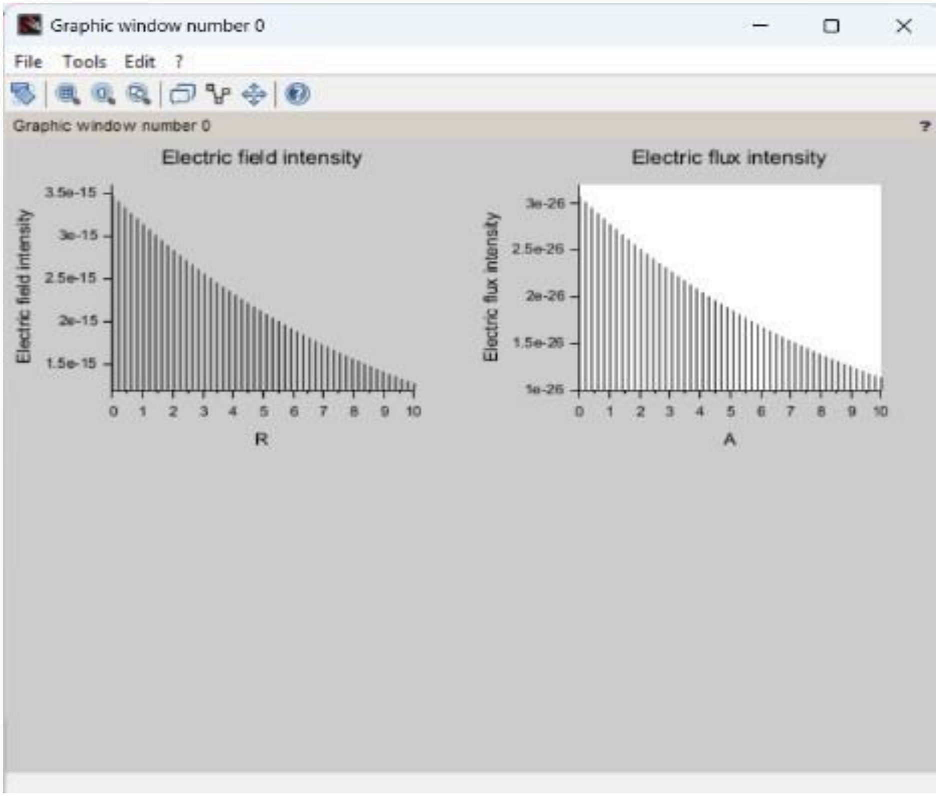
•SCILAB version 6.0.1 **Code**:

EXP 1 EME.sci | EXp 1)ii.sci | exp 1)iii eme.sci | exp 2.sci | exp 2 case 1.sci

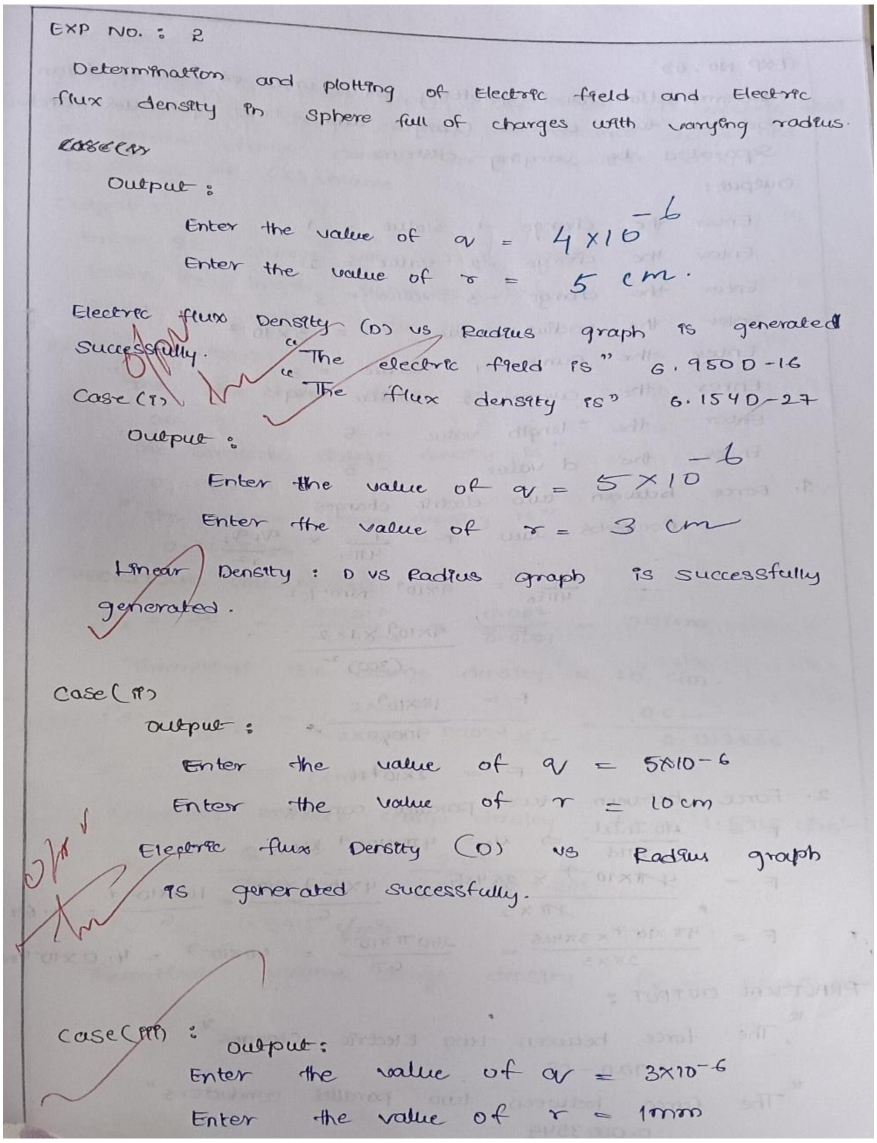
```
clc;
clear;

q=input('Enter the value of q=');
r=input('Enter the value of r=');

efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]); x = linspace(0, 10, 50);
y = E * exp(-0.1 * x);
z = D * exp(-0.1 * x);

figure();
subplot(2, 2, 1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

Output:


Scilab 2026.1.0 Console
File Edit Control Applications ?
File Browser
C:\Users\user\Documents\
Name
Documents
My Music
My Pictures
My Videos
ECA05 EXP 2.sci~
ex 6.sci
Scilab 2026.1.0 Console
--> exec('C:\Users\user\OneDrive\Documents\EX-2(I).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EX-2(I).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\EXP-2(1).sci', -1)
--> exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci', -1)
Enter the value of q = exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci',
Warning: file 'C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci' already openec
Enter the value of q = 10
Enter the value of r = exec('C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci',
Warning: file 'C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci' already openec
Enter the value of q = 5*10-6
Enter the value of r = 10
"The electric field intensity is:"
3.955D+09
"E = "
"The electric flux density is:"
0.0350141
"D = "


Graphic window number 0
File Tools Edit ?
Graphic window number 0
Electric field intensity
Electric field intensity
3.5e-15
3e-15
2.5e-15
2e-15
1.5e-15
0 1 2 3 4 5 6 7 8 9 10
R
Electric flux intensity
Electric flux intensity
3e-26
2.5e-26
2e-26
1.5e-26
1e-26
0 1 2 3 4 5 6 7 8 9 10
A

Theoretical calculation:



EXP NO. : 2

Determination and plotting of Electric field and Electric flux density in Sphere full of charges with varying radius.

~~case(i)~~

Output :

Enter the value of q = 4×10^{-6}

Enter the value of r = 5 cm.

Electric flux Density (D) vs Radius graph is generated successfully.

"The electric field is" 6.950D-16

"The flux density is" 6.154D-27

Case (i)

Output :

Enter the value of q = 5×10^{-6}

Enter the value of r = 3 cm

Linear Density : D vs Radius graph is successfully generated.

Case (ii)

output :

Enter the value of q = 5x10-6

Enter the value of r = 10 cm

Electric flux Density (D) vs Radius graph is generated successfully.

case (iii) :

output :

Enter the value of q = 3x10-6

Enter the value of r = 1mm

Case 1:

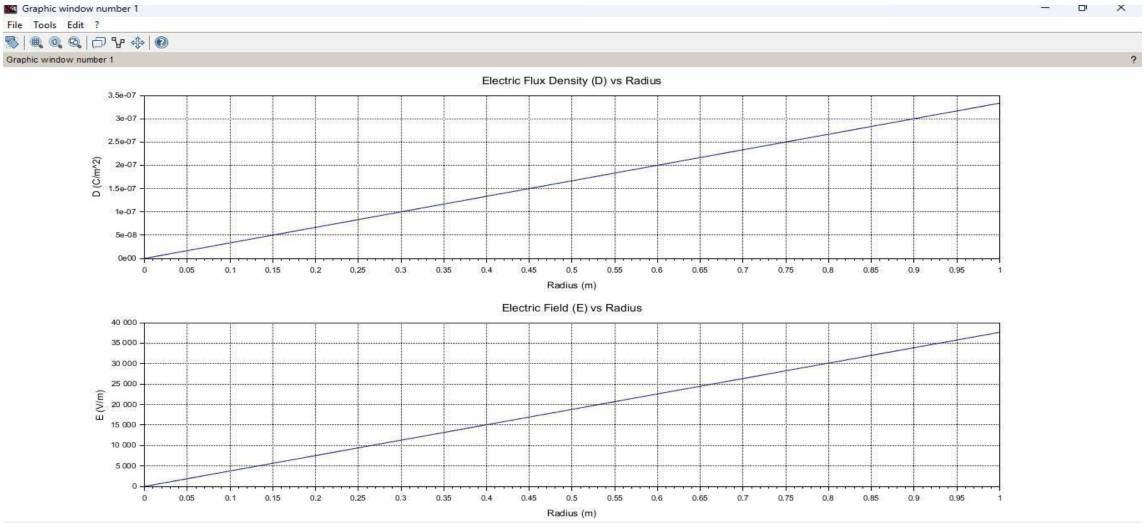
Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ0. What are the values of E and D at the center and at the surface of the sphere?

Code:

ECA05 EXP 2.sci (C:\Users\user\OneDrive\Documents\ECA05 EXP 2.sci) - SciNotes

```
clc;
clear;

// Uniform charge density
rho0 = 1e-6;               // Charge density (C/m^3)

// Permittivity of free space
epsilon0 = 8.854e-12;      // F/m

// Radius of the sphere
R = 1;                     // m

// Number of points
n = 100;

// Radial distance
r = linspace(0, R, n);

// Initialise arrays
D = zeros(1, n);
E = zeros(1, n);

// Calculate D and E inside the uniformly charged sphere
for i = 1:n

    ri = r(i);

    if ri == 0 then
        D(i) = 0;
        E(i) = 0;
    else
        // Electric flux density
        D(i) = (rho0 * ri) / 3;

        // Electric field intensity
        E(i) = D(i) / epsilon0;
    end

end

// Create figure
scf(0);
clf();

// Plot D vs Radius
subplot(1, 2, 1);
plot(r, D);
xlabel("Radius (m)");
ylabel("Electric Flux Density D (C/m^2)");
title("Uniform Charge Density: D vs Radius");
xgrid();

// Plot E vs Radius
subplot(1, 2, 2);
plot(r, E);
xlabel("Radius (m)");
ylabel("Electric Field E (V/m)");
title("Uniform Charge Density: E vs Radius");
xgrid();
```

Output:



Case 2:

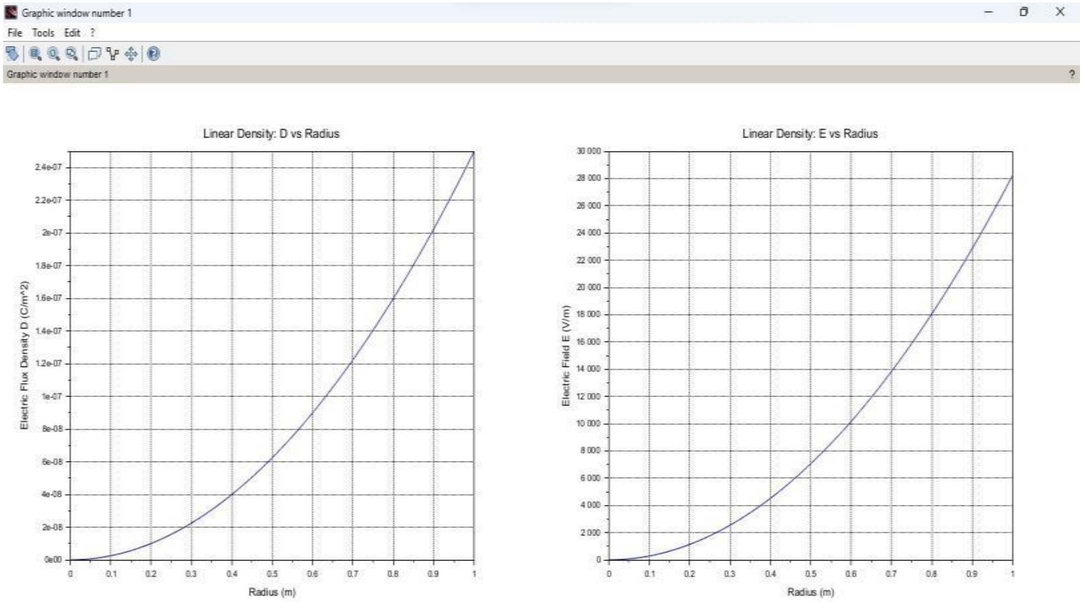
For a sphere with a linearly increasing charge density ρv=ρ0r.derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.

Code:

```
clc;
clear;

// Parameters
rho0 = 1e-6;                // Coefficient of charge density (C/m^4)
epsilon0 = 8.854e-12;       // Permittivity of free space (F/m)
R = 1;                      // Radius of sphere (m)
n = 100;                    // Number of points

// Radial distances
r = linspace(0, R, n);

// Initialise arrays
D = zeros(1, n);
E = zeros(1, n);

// Calculate D and E
for i = 1:n

    ri = r(i);

    if ri == 0 then
        D(i) = 0;
        E(i) = 0;
    else
        // Enclosed charge
        // Q_enclosed = rho0 * pi * ri^4
        Q_enclosed = rho0 * %pi * ri^4;

        // Electric flux density
        D(i) = Q_enclosed / (4 * %pi * ri^2);

        // Electric field intensity
        E(i) = D(i) / epsilon0;
    end

end

// Create figure
scf(0);
clf();

// Plot D vs Radius
subplot(1, 2, 1);
plot(r, D);
xlabel("Radius (m)");
ylabel("Electric Flux Density D (C/m^2)");
title("Linearly Increasing Density: D vs Radius");
xgrid();

// Plot E vs Radius
subplot(1, 2, 2);
plot(r, E);
xlabel("Radius (m)");
ylabel("Electric Field E (V/m)");
title("Linearly Increasing Density: E vs Radius");
xgrid();
```

Backup finished...

output:

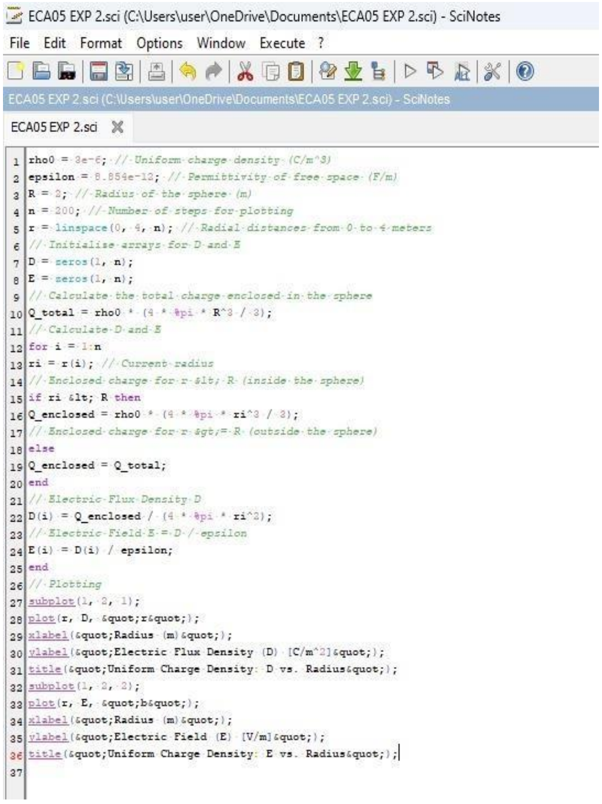


Case 3:

A sphere of radius R=2 m has a uniform charge density ρ=3×10^−6 C/m^3

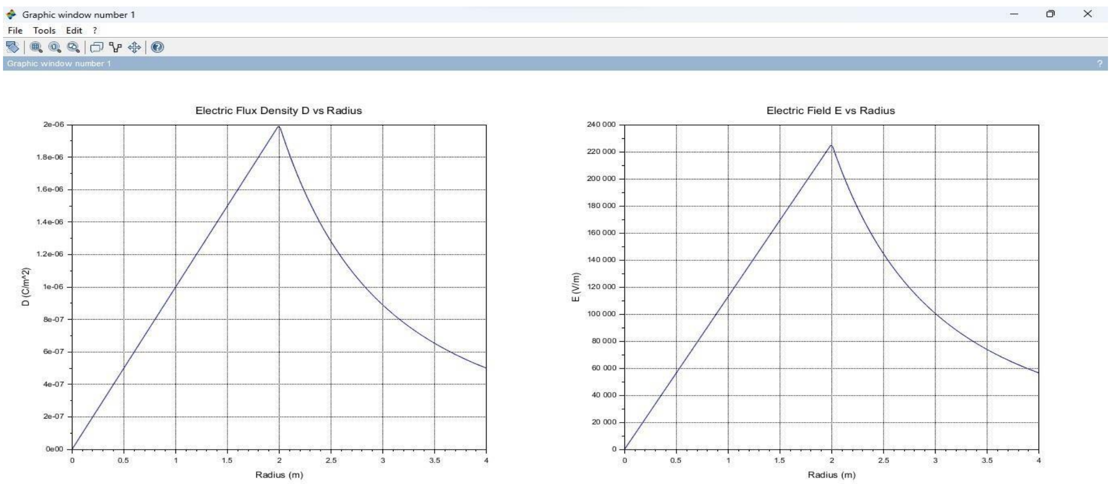
1. **Calculate and plot the electric flux density D(r) and electric field E(r) as a function of radial distance r, ranging from 0 to 4 meters.**
2. **Discuss the behaviour of the electric field inside and outside the sphere.**

Code:



```
rho0 = 3e-6; // Uniform charge density (C/m^3)
epsilon = 8.854e-12; // Permittivity of free space (F/m)
R = 2; // Radius of the sphere (m)
n = 200; // Number of steps for plotting
r = linspace(0, 4, n); // Radial distances from 0 to 4 meters
// Initialise arrays for D and E
D = zeros(1, n);
E = zeros(1, n);
// Calculate the total charge enclosed in the sphere
Q_total = rho0 * (4 * %pi * R^3 / 3);
// Calculate D and E
for i = 1:n
ri = r(i); // Current radius
// Enclosed charge for r &lt; R (inside the sphere)
if ri &lt; R then
Q_enclosed = rho0 * (4 * %pi * ri^3 / 3);
// Enclosed charge for r &gt;= R (outside the sphere)
else
Q_enclosed = Q_total;
end
// Electric Flux Density D
D(i) = Q_enclosed / (4 * %pi * ri^2);
// Electric Field E = D / epsilon
E(i) = D(i) / epsilon;
end
// Plotting
subplot(1, 2, 1);
plot(r, D, &quot;r&quot;);
xlabel(&quot;Radius (m)&quot;);
ylabel(&quot;Electric Flux Density (D) [C/m^2]&quot;);
title(&quot;Uniform Charge Density: D vs. Radius&quot;);
subplot(1, 2, 2);
plot(r, E, &quot;b&quot;);
xlabel(&quot;Radius (m)&quot;);
ylabel(&quot;Electric Field (E) [V/m]&quot;);
title(&quot;Uniform Charge Density: E vs. Radius&quot;);
```

output:



Result :

Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.